EXPERIMENT – 7

Secure Wi-Fi Configuration using Windows

Aim

To configure a wireless router securely by changing default settings, configuring neutral SSIDs, enabling WPA2/WPA3 security, disabling WPS, and verifying authentication and encryption parameters using Windows CLI commands.

Requirements

- Wireless router / 4G router or Wi-Fi simulator
- Windows 10/11 computer with Wi-Fi adapter
- Web browser (Chrome, Edge)
- Command Prompt (cmd)
- Authorized router administrator credentials

> **Safety Warning:** Perform this experiment only on an authorized router or network simulator. If using a physical 4G router, do not modify mobile APN or SIM settings. Maintain an active wired connection or keep default credentials accessible to avoid lockouts.

Procedure

1. Identify Router IP Address

Open Command Prompt and execute:

```
ipconfig
```

Locate the Default Gateway address in the output.

Expected Output:

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : Home
   IPv4 Address. . . . . . . . . . . : 192.168.8.100
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.8.1
```

2. Access Router Configuration Page

Open a web browser and enter the Default Gateway IP into the address bar:
`http://192.168.8.1`
Log in using authorized administrator credentials.

3. Configure Administrator Password

Navigate to **System / Management Settings** and change the factory default administrator password. Set a strong password that is strictly different from the Wi-Fi passphrase.

4. Change the Wireless Network Name (SSID)

1. Open Wireless > Basic Settings.
2. Change the default SSID to a neutral name such as CampusLab-Secure.
3. For a dual-band router, configure both 2.4 GHz and 5 GHz SSIDs. They may use the same name or separate names such as CampusLab-Secure-5G.
4. Do not include a person's full name, address, phone number, organisation password, or router model in the SSID.
5. Keep SSID broadcast enabled unless the laboratory specifically requires otherwise. Hiding an SSID is not a strong security control.

5. Configure WPA2/WPA3 Security & Encryption

Navigate to **Wireless > Security Settings**. Select **WPA2/WPA3-Personal** as the security mode, choose **AES / CCMP** encryption, and set a strong Wi-Fi passphrase.

6. Disable WPS (Wi-Fi Protected Setup)

1. Open Wireless > WPS, Advanced Wireless, or Wi-Fi Protected Setup.
2. Set WPS to Disabled.
3. Confirm that both the WPS PIN method and push-button method are off, when separate controls are shown.
4. Click Save.

Summary of Wireless Security Configurations

Setting	Value	Meaning
SSID Name	CampusLab-Secure	Neutral identifier without sensitive information
Security Mode	WPA2/WPA3-Personal	Secure authentication protocol
Encryption	AES / CCMP	High-level robust data encryption
WPS Status	Disabled (PIN & Push-button)	Prevents brute-force PIN attacks

Save the settings and allow the router to reboot wireless services.

7. Verify Hardware WPA2/WPA3 & AES Support

Verify that the Wi-Fi network interface driver supports WPA2-Personal, WPA3-Personal, and AES (CCMP) encryption:

```
netsh wlan show drivers
```

Expected Output:

```
Interface name: Wi-Fi

    Driver                   : Intel(R) Wi-Fi 6 AX201 160MHz
    Provider                 : Intel
    Authentication and cipher supported in infrastructure mode:
        Open             None
        Open             WEP-40
        WPA2-Personal    CCMP
        WPA3-Personal    CCMP
        WPA2-Enterprise  CCMP
    Management Frame Protection supported : Yes
```

8. Scan & Verify Available Wireless Networks

Scan for broadcast networks to verify that the router is broadcasting the neutral SSID with WPA2/WPA3 authentication and CCMP cipher:

```
netsh wlan show networks mode=bssid
```

Expected Output:

```
Interface name : Wi-Fi
There are 1 networks currently available.

SSID 1 : CampusLab-Secure
    Network type            : Infrastructure
    Authentication          : WPA2-Personal
    Encryption              : CCMP
    BSSID 1                 : e8:48:b8:12:34:56
         Signal             : 98%
         Radio type         : 802.11ax
         Channel            : 6
```

9. Inspect Saved Profile Security Settings

Connect the Windows PC to CampusLab-Secure, then inspect the saved profile's authentication mode, cipher, and security key status:

```
netsh wlan show profile name="CampusLab-Secure" key=clear
```

Expected Output:

```
Profile CampusLab-Secure on interface Wi-Fi:
=======================================================================
Profile Settings
----------------
    Version                : 1
    Type                   : Wireless LAN
    Name                   : CampusLab-Secure
    Control options        : Connect automatically
Security settings
-----------------
    Authentication         : WPA2-Personal
    Cipher                 : CCMP
    Security key           : Present
    Key Content            : SecurePass123!
```

10. Verify Active Connection Parameters

Verify active real-time connection status and encryption parameters:

```
netsh wlan show interfaces
```

Expected Output:

```
There is 1 interface on the system:

    Name                   : Wi-Fi
    Description            : Intel(R) Wi-Fi 6 AX201 160MHz
    State                  : connected
    SSID                   : CampusLab-Secure
    BSSID                  : e8:48:b8:12:34:56
    Network type           : Infrastructure
    Radio type             : 802.11ax
    Authentication         : WPA2-Personal
    Cipher                 : CCMP
    Channel                : 6
    Receive rate (Mbps)    : 574
    Transmit rate (Mbps)   : 574
    Signal                 : 98%
```

Verification

Test / Command	Expected Result	Status
ipconfig	Default Gateway IP displayed	**Pass**
Admin Password Change	Default admin password updated securely	**Pass**
SSID Configuration	Neutral SSID configured & broadcast enabled	**Pass**
WPS Status Check	PIN and Push-button WPS disabled in router portal	**Pass**
netsh wlan show drivers	WPA2/WPA3 and CCMP (AES) supported	**Pass**
netsh wlan show interfaces	State connected with CCMP cipher	**Pass**

Precautions

- Perform the experiment only on authorized routers or network simulators.
- Do not include personal information, phone numbers, or router models in the SSID.
- Keep SSID broadcast enabled unless specifically required otherwise.
- Ensure both PIN and push-button WPS methods are completely disabled.
- Do not use obsolete or weak security protocols such as WEP, WPA, or TKIP.
- Keep administrator and Wi-Fi passwords strictly different.
- Do not interrupt router power during saving settings or rebooting.
- If using a physical 4G router, do not modify APN or mobile SIM data settings.

Result

The wireless router was successfully configured with a neutral SSID, secure authentication (WPA2/WPA3), AES encryption, disabled WPS (PIN and push-button methods), and verified using Windows network diagnostic commands.

The experiment is successful if:

- Router IP address is identified using ipconfig.
- Default administrator password is changed to a strong password.
- SSID is set to a neutral name (e.g., CampusLab-Secure) without personal information.
- SSID broadcast remains enabled as per security best practices.
- Both WPS PIN and push-button methods are set to Disabled.
- Security mode is set to WPA2/WPA3-Personal with AES / CCMP encryption.
- Active connection status and CCMP cipher are confirmed via netsh wlan show interfaces.